

Terraform Task

Task Description:

Launch Linux EC2 instances in two regions using a single Terraform file.

Step 1 : Configured aws cli in terraform server.

```
ubuntu@ip-172-31-40-191:~$ aws configure
AWS Access Key ID [None]: AKIA4F35FI6UL3H0DB60
AWS Secret Access Key [None]: /5Hoewbtd3FtkUR/vLbMkmICmkM622qckkwhYVbJ
Default region name [None]: us-west-2
Default output format [None]:
```

Step 2 - Created a terraform script to launch instances in N. Verginia and Mumbai

```
terraform {

  required_providers {

    aws = {

      source = "hashicorp/aws"

      version = "~> 5.0"

    }

  }

}

provider "aws" {

  alias = "us_east_1"

  region = "us-east-1"
```

```

}

provider "aws" {

  alias = "ap_south_1"

  region = "ap-south-1"
}

data "aws_ami" "amazon_linux_us" {

  provider = aws.us_east_1

  most_recent = true

  owners = ["amazon"]

  filter {

    name = "name"

    values = ["al2023-ami-*-x86_64"]

  }

}

data "aws_ami" "amazon_linux_in" {

  provider = aws.ap_south_1

  most_recent = true

  owners = ["amazon"]

  filter {

    name = "name"

    values = ["al2023-ami-*-x86_64"]

  }

}

resource "aws_instance" "us_ec2" {

  provider = aws.us_east_1

  ami = data.aws_ami.amazon_linux_us.id

```

```
instance_type = "t2.micro"
```

```
tags = {
```

```
    Name = "linux-us-east-1"
```

```
}
```

```
}
```

```
resource "aws_instance" "in_ec2" {
```

```
    provider      = aws.ap_south_1
```

```
    ami           = data.aws_ami.amazon_linux_in.id
```

```
    instance_type = "t2.micro"
```

```
    tags = {
```

```
        Name = "linux-ap-south-1"
```

```
    }
```

```
}
```

Step 3 - Executed the script

terraform init

```
ubuntu@ip-172-31-40-191:~/project$ terraform init
Initializing the backend...
Initializing provider plugins...
- Finding hashicorp/aws versions matching "~> 5.0"...
- Installing hashicorp/aws v5.100.0...
- Installed hashicorp/aws v5.100.0 (signed by HashiCorp)
Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.
```

terraform plan

```
ubuntu@ip-172-31-40-191:~/project$ terraform plan
data.aws_ami.amazon_linux_us: Reading...
data.aws_ami.amazon_linux_in: Reading...
data.aws_ami.amazon_linux_us: Read complete after 1s [id=ami-02f058bc6b3ee6fcc]
data.aws_ami.amazon_linux_in: Read complete after 2s [id=ami-02a8bcaaf6d87b759]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

Terraform will perform the following actions:

# aws_instance.in_ec2 will be created
+ resource "aws_instance" "in_ec2" {
  + ami              = "ami-02a8bcaaf6d87b759"
  + arn              = (known after apply)
  + associate_public_ip_address = (known after apply)
}
```

terraform apply

```
ubuntu@ip-172-31-40-191:~/project$ terraform apply
data.aws_ami.amazon_linux_us: Reading...
data.aws_ami.amazon_linux_in: Reading...
data.aws_ami.amazon_linux_us: Read complete after 1s [id=ami-02f058bc6b3ee6fcc]
data.aws_ami.amazon_linux_in: Read complete after 2s [id=ami-02a8bcaaf6d87b759]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

Terraform will perform the following actions:

# aws_instance.in_ec2 will be created
+ resource "aws_instance" "in_ec2" {
  + ami              = "ami-02a8bcaaf6d87b759"
  + arn              = (known after apply)
}
```

Do you want to perform these actions?

Terraform will perform the actions described above.
Only 'yes' will be accepted to approve.

Enter a value: yes

```
aws_instance.us_ec2: Creating...
aws_instance.in_ec2: Creating...
aws_instance.us_ec2: Still creating... [00m10s elapsed]
aws_instance.in_ec2: Still creating... [00m10s elapsed]
aws_instance.us_ec2: Creation complete after 14s [id=i-0229e4cec05fb7ac5]
aws_instance.in_ec2: Creation complete after 15s [id=i-04feccd7de1c5ee2e]
```

Apply complete! Resources: 2 added, 0 changed, 0 destroyed.

Instances started running in two regions

Instance summary for i-0229e4cec05fb7ac5 (linux-**us-east-1**)

Updated less than a minute ago

Instance ID

i-0229e4cec05fb7ac5

IPv6 address

–

Hostname type

IP name: ip-172-31-24-176.ec2.internal

Answer private resource DNS name

–

Auto-assigned IP address

3.94.146.197 [Public IP]

IAM role

–

IMDSv2

Required

Operator

–

Public IPv4 address

3.94.146.197 | [open address](#)

Instance state

Running

Private IP DNS name (IPv4 only)

ip-172-31-24-176.ec2.internal

Instance type

t3.micro

VPC ID

vpc-03e1852553eb9dd75

Subnet ID

subnet-02e8cc4a115b24ff6

Instance ARN

arn:aws:ec2:us-east-1:837244241832:instance/i-0229e4cec05fb7ac5

Private IPv4 addresses

172.31.24.176

Public DNS

ec2-3-94-146-197.compute-1.amazonaws.com | [open address](#)

Elastic IP addresses

–

AWS Compute Optimizer finding

Opt-in to AWS Compute Optimizer for recommendations. | [Learn more](#)

Auto Scaling Group name

–

Managed

false

Connect

Instance state ▼

Actions ▼

Instance summary for i-04feccd7de1c5ee2e (linux-ap-south-1)

Updated less than a minute ago

Instance ID

i-04feccd7de1c5ee2e

IPv6 address

–

Hostname type

IP name: ip-172-31-46-4.ap-south-1.compute.internal

Answer private resource DNS name

–

Auto-assigned IP address

13.127.186.38 [Public IP]

IAM role

–

IMDSv2

Required

Operator

–

Public IPv4 address

13.127.186.38 | [open address](#)

Instance state

Running

Private IP DNS name (IPv4 only)

ip-172-31-46-4.ap-south-1.compute.internal

Instance type

t3.micro

VPC ID

vpc-00f1b106c0d708742

Subnet ID

subnet-003ad8b0f91fd320d

Instance ARN

arn:aws:ec2:ap-south-1:837244241832:instance/i-04feccd7de1c5ee2e

Private IPv4 addresses

172.31.46.4

Public DNS

ec2-13-127-186-38.ap-south-1.compute.amazonaws.com | [open address](#)

Elastic IP addresses

–

AWS Compute Optimizer finding

Opt-in to AWS Compute Optimizer for recommendations. | [Learn more](#)

Auto Scaling Group name

–

Managed

false

Connect

Instance state ▼

Actions ▼